Armored Instrumentation Cable
UL Type MC HL, 600V, 90°C rated - Lead Free

Single or multiple individually shielded pairs or triads, overall cable shield, continuous corrugated aluminum sheath, PVC jacket

Construction

Conductor: bare, annealed copper conforming to ASTM B3 and Class B stranded in accordance with ASTM B8.

Insulation: PVC/Nylon type TFN in accordance with UL 66, flame retardant, 90°C temperature rating.

Insulation shield: aluminum foil/polyester shield helically wrapped to provide 100% coverage and tinned copper drain wire that is two gauge sizes smaller than the circuit conductors. These shields are electrically isolated from each other.

Assembly: pairs/triads are cabled in concentric layers with interstices filled with suitable non-hygroscopic fillers, as required. A binder tape of synthetic material assembles the core in an essentially round configuration.

Overall cable shield: aluminum foil/polyester shield helically wrapped to provide 100% coverage and tinned copper drain wire that is the same size as the circuit conductors.

Inner jacket: polyvinyl chloride jacket, over cabled core as per UL 1569, 90°C temperature rating, with additional resistance to fire spread. A rip cord is laid longitudinally under the jacket to facilitate stripping.

Armor: continuous corrugated aluminum sheath with no more than 0.2% trace copper providing complete protection against liquid and gas ingress. It also provides excellent mechanical protection, additional electrostatic shielding, and serves as an easy means for grounding equipment.

Jacket: overall black polyvinyl chloride jacket per UL 1569, 90°C temperature rating; low acid gas emission; limited flame spread and excellent corrosion resistance.

Conductor Identification

Pairs: black/white & number coded
Triads: black/white/red & number coded

Bending Radius

Fixed position: 7 x cable overall diameter
During pulling: 12 x cable overall diameter

Specifications

- Meets UL 66, TFN rated 90°C 600V conductors
- Meets UL 1569 requirements for Type MC, Metal Clad cables
- Meets UL 2225 for Hazardous Locations
- Designated Type MC as per NEC Article 330

Product Features

- UL approved cables Type MC, 600V; File No. E47409
- UL approved insulated conductors
- Cables pass UL 1685 and IEEE 383 vertical tray fire tests at 70,000 BTU/hr, ICEA T-29-520 fire test at 210,000 BTU/hr, IEC 332-3 category A fire test, IEEE 1202 and CSA FT4
- Cables are American Bureau of Shipping (ABS) listed as CWC MC Type MC
- Cables exhibit a low temperature rating of in excess of –40°C impact and –40°C bend with suitable precautions
- Continuous, impervious aluminum sheath corrugated for flexibility, prevents ingress of moisture, gases and liquids
- Aluminum sheath cross-section exceeds requirements of the NEC Section 250.178 for grounding conductor
- Excellent mechanical and physical properties
- Minimal noise and signal interference
- Sunlight resistant jacket
- Suitable for direct burial, use in cable tray and embedment in concrete
- Lead free and RoHS compliant

Options

The following constructions can be provided on special orders:

- Different conductor size
- Different pair or triad configurations
- Specially colored jackets
- Other constructions and combinations (some manufacturing restrictions apply)
- Oil Resistant I or II jackets
- UL 1309 listing and marking

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Multi Pairs, 600V – 18 AWG (7w) • Insulation Thickness: 15 mils / 0.38 mm PVC and 4 mils / 0.10mm Nylon

# of Pairs	Nominal Diameter over Core		Inner Jacket Thickness		Nominal Diameter over Inner Jacket		Nominal Diameter over Sheath		Outer Jacket Thickness		Nominal Diameter over Outer Jacket		Approximate Net Cable Weight	
	inches	mm	mils	mm	inches	mm	inches	mm	mils	mm	inches	mm	lb/kft	kg/km
2	.324	8.23	40	1.02	.410	10.41	.578	14.68	50	1.27	.702	17.83	222	330
4	.400	10.16	40	1.02	.485	12.32	.648	16.46	50	1.27	.753	19.13	268	399
8	.502	12.75	50	1.27	.607	15.42	.815	20.70	50	1.27	.920	23.37	420	625
12	.680	17.27	50	1.27	.785	19.94	1.000	25.40	50	1.27	1.105	28.07	560	833
16	.703	17.86	50	1.27	.807	20.50	1.124	28.55	50	1.27	1.226	31.14	706	1051
24	.980	24.89	50	1.27	1.084	27.53	1.392	35.36	50	1.27	1.493	37.92	969	1442

Multi Pairs, 600V – 16 AWG (7w) • Insulation Thickness: 15 mils / 0.38 mm PVC and 4 mils / 0.10mm Nylon

# of Pairs	Nominal Diameter over Core		Inner Jacket Thickness		Nominal Diameter over Inner Jacket		Nominal Diameter over Sheath		Outer Jacket Thickness		Nominal Diameter over Outer Jacket		Approximate Net Cable Weight	
	inches	mm	mils	mm	inches	mm	inches	mm	mils	mm	inches	mm	lb/kft	kg/km
1	.204	5.18	40	1.02	.285	7.24	.482	12.24	50	1.27	.584	14.83	160	238
2	.424	10.77	40	1.02	.506	12.85	.752	19.10	50	1.27	.856	21.74	246	366
4	.482	12.24	50	1.27	.584	14.83	.797	20.24	50	1.27	.901	22.89	315	469
8	.635	16.13	50	1.27	.740	18.80	.968	24.59	50	1.27	1.072	27.23	564	839
12	.797	20.24	50	1.27	.900	22.86	1.190	30.23	50	1.27	1.294	32.87	761	1133
16	.882	22.40	50	1.27	.986	25.04	1.325	33.66	50	1.27	1.433	36.40	958	1426
24	1.124	28.55	50	1.27	1.228	31.19	1.570	39.88	60	1.52	1.693	43.00	1345	2002
36	1.312	33.32	50	1.27	1.416	35.97	1.740	44.20	60	1.52	1.862	47.29	1760	2619

Multi Triads, 600V – 16 AWG (7w) • Insulation Thickness: 15 mils / 0.38 mm PVC and 4 mils / 0.10mm Nylon

# of Triads	Nominal Diameter over Core		Inner Jacket Thickness		Nominal Diameter over Inner Jacket		Nominal Diameter over Sheath		Outer Jacket Thickness		Nominal Diameter over Outer Jacket		Approximate Net Cable Weight	
	inches	mm	mils	mm	inches	mm	inches	mm	mils	mm	inches	mm	lb/kft	kg/km
1	.225	5.72	40	1.02	.310	7.87	.482	12.24	50	1.27	.585	14.86	157	232
4	.500	12.70	50	1.27	.604	15.34	.808	20.52	50	1.27	.910	23.11	426	634
8	.590	14.99	50	1.27	.694	17.63	.940	23.88	50	1.27	1.044	26.52	639	951
12	.865	21.97	50	1.27	.969	24.61	1.315	33.40	50	1.27	1.423	36.14	962	1432